MATH 112 — Practice Exam 2B

Open notes **Suggested time: 80 minutes** **Total: 100 points**

Name: _____

Instructions. Attempt all seven problems. Show your work and justify your answers except where explicitly stated otherwise.

Problem 1. True or False (16 points)

For each statement, circle **T** or **F**. Each part is worth **2 points**. **No justification is required.**

(a) If for all $\varepsilon > 0$ there exist x, y such that $|x - y| < \varepsilon$, then $x = y$. T F

(b) If a set A has a supremum, then it has a maximum and $\max(A) = \sup(A)$. T F

(c) Let A be a nonempty subset of $\mathbb{R}$ that is upper bounded.
For every $x \in A$, we have $x < \sup A$. T F

(d) Every closed ball in a metric space is also an open set. T F

(e) Every open set in a metric space is an open ball. T F

(f) The intersection of any collection of open sets is open. T F

(g) For every $z, w \in \mathbb{C}$,

$$|z + w| = |z| + |w|.$$

T F

(h) Absolute value defines a norm on $\mathbb{R}$. T F

Problem 2 (14 points)

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above.

(a) If $A \cap B \neq \varnothing$, prove

$$\sup(A \cap B) \leq \min\{\sup A, \sup B\}.$$

(5 points)

(b) Suppose there is a constant $c > 0$ such that

$$|a - b| \leq c \qquad \text{for every } a \in A \text{ and } b \in B.$$

Prove that $|\sup A - \sup B| \leq c$. (9 points)

Problem 3 (14 points)

This problem is on the harder side, but it is good practice for connecting different topics.

(a) Let $a, b \in \mathbb{R}$ with $a < b$. Using density of rational numbers and the irrationality of $\sqrt{2}$, prove that some irrational number lies strictly between a and b. (6 points)

(b) Using the density of rationals and part (a), prove that neither $\mathbb{Q}$ nor $\mathbb{R} \setminus \mathbb{Q}$ is open or closed in $\mathbb{R}$ with its usual metric. (8 points)

Problem 4 (14 points)

(a) Find all complex numbers z satisfying

$$z + \overline{z} = 6, \qquad |z| = 5.$$

Write your answers in the form $a + bi$. (3 points)

(b) Prove that $|z + \overline{z}| \leq 2|z|$ for every $z \in \mathbb{C}$. For which z does equality hold? (3 points)

(c) Find all nonzero complex numbers z such that

$$\frac{z}{\overline{z}} = -1.$$

Justify your answer. (3 points)

(d) Find a fourth root of

$$-\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

in $\mathbb{C}$, i.e., find $z = a + bi$ such that $z^4 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$. Specify the real numbers a and b. *Hint:* Write z^4 in polar form first. What is the polar form of z? (5 points)

The table below might be useful:

x	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π
$\sin x$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0
$\cos x$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0	$-1/2$	$-\sqrt{2}/2$	$-\sqrt{3}/2$	-1

Problem 5(14 points)

(a) Describe geometrically the set $S = \{z \in \mathbb{C} : |z - 1| = |z + 1|\}$. Find all points $z \in S$ satisfying $|z| = 2$. (7 points)

(b) On $\mathbb{R}$, consider the discrete metric $d_{\text{disc}}(x, y) = 0$ if $x = y$ and 1 otherwise. Describe $B_{d_{\text{disc}}}(2, 1)$ and $D_{d_{\text{disc}}}(2, 1)$. (4 points)

(c) Explain why the open ball $B(0, 1)$ in $\mathbb{R}$ with the usual distance is an open set. (3 points)

Problem 6 (14 points)

Let (a_n) be a sequence of positive real numbers. Prove, using the definitions, that

$$a_n \longrightarrow 0 \quad \Longleftrightarrow \quad \frac{1}{a_n} \longrightarrow +\infty.$$

(14 points)

Problem 7 (14 points)

(a) Using the ε–$n_0(\varepsilon)$ definition of convergence, prove that

$$a_n = \frac{n+3}{n+5} \longrightarrow 1.$$

Give a valid choice of $n_0(\varepsilon)$. (4 points)

(b) Evaluate

$$\lim_{n\to\infty} \left(\sqrt{n^2+3n} - n\right),$$

and justify your answer. (3 points)

(c) Suppose $a_n \to 0$ and $a_n b_n \to 0$. Must (b_n) be bounded? Justify your answer with a proof or counterexample. (3 points)

(d) Suppose (s_n) is a convergent sequence of complex numbers and $s_n \to s$. Prove that the real sequence $(\mathrm{Re}(s_n))$ converges and that

$$\lim_{n\to\infty} \mathrm{Re}(s_n) = \mathrm{Re}(s).$$

(4 points)